Kingdom of Saudi Arabia Ministry of Education University of Jeddah College of Computer Science and Engineering Department of Computer Science and Artificial Intelligence	جامعة جدة University of Jeddah	المملكة العربية السعودية وزارة التعليم جامعة جدّة كلية علوم وهندسة الحاسب قسم علوم الحاسب والذكاء الاصطناعي

Project: CCAI 435

Marks: 10

Due date: 14 May 2026

PLO/CLO
PLO S2 (CLO 2.2): Implement deep models for real-world problems using state of-the-art tools, techniques and practices
PLO C4 (CLO 3.1) : Discover and apply new knowledge as needed, using appropriate learning strategies
PLO C1 (CLO 3.2): Demonstrate competence related to presenting and communicating ideas effectively to solve problems related to deep learning

1. Project Purpose

This project constitutes the **final practical assessment** of the Deep Learning course.
The objective of the project is to **identify a real-world issue** in **Education**, **Healthcare**, or **Logistics** and to **develop a functional prototype as a deep learning–based solution**.

Students are required to move beyond theoretical modeling and demonstrate how deep learning can be used to **tackle practical problems through an implemented prototype**.

The project assesses students' ability to:

- Analyze real-world issues
- Design an appropriate deep learning solution
- Integrate multiple system components
- Evaluate system performance
- Deploy a usable prototype with a user interface

2. Group Formation

- The project must be completed in **groups of exactly three (3) students**.
- Groups with fewer or more than three members will **not be accepted**.
- All members are expected to contribute equally.
- During evaluation, any group member may be asked to explain any part of the project.

3. Project Design Options (Select ONE)

Each group must select **one** of the following system designs.
The selected design must be clearly justified and consistently implemented throughout the project.

Option A: Hybrid Deep Learning System

Groups selecting this option are required to design a system composed of **multiple cooperating deep learning models** to address the selected issue.

Requirements

- Implement **at least three (3) deep learning models**.
- Integrate all models into a **single unified system** that produces one final output.
- Ensure that the system functions as a **working prototype**, not as isolated experiments.

Option B: Retrieval-Augmented Generation (RAG) System

Groups selecting this option are required to develop a **prototype system** that answers user queries by retrieving and generating information from documents related to the chosen issue.

Requirements

- Collect a set of **domain-specific documents** relevant to the identified problem.

- Preprocess documents and store them as embeddings.
- Implement a retrieval mechanism to select relevant documents for each query.
- Use a generative model to produce responses **grounded in retrieved content**.
- Ensure the system operates as a **working prototype** that demonstrates the RAG concept end-to-end.

4. System Interface and Deployment (Mandatory)

All projects must be delivered as **deployed prototypes** that users can interact with.

Interface Requirements

- The prototype must include a **user interface**.
- Users must be able to:
    - Provide input (e.g., text, queries, or data)
    - Trigger system processing
    - View outputs clearly

The interface does not need advanced design; the focus is on **usability and functionality**.

Deployment Requirements

- The prototype must be deployed as a **working application**, either locally or online.
- Acceptable deployment frameworks include:
    - Streamlit
    - Flask
    - FastAPI
- The deployed prototype must demonstrate complete system flow (input → processing → output).

Deployment Evidence

Groups must submit:

- Screenshots or a short demonstration video of the deployed prototype
- Examples of user input and system output

5. Problem Domain Selection

Each group must select **one real-world issue** within one of the following domains and justify its importance.

Education

Healthcare

Logistics

6. Technical Requirements

All projects must adhere to the following technical constraints:

- Dataset: **Publicly available or instructor-approved**
- Evaluation metrics must be appropriate to the task and clearly reported

7. Final Submission Requirements

Each group must submit the following:

1. Project Report

The report must include:

- Description of the identified issue
- Proposed solution and system architecture
- Description of models or RAG components
- Dataset description

- Evaluation methodology and results
- Prototype interface and deployment description

2. Source Code

- Well-organized and documented code
- Clear instructions for running the prototype

3. Interface and Deployment Evidence

- Screenshots or short demo video of the deployed prototype
- Optional live deployment link

**** Evaluation Criteria**

Projects will be evaluated based on:

- Clarity of the problem and proposed solution
- System architecture and design
- Model integration or RAG implementation
- Quality of the prototype interface and deployment
- Evaluation and analysis
- Quality of report and presentation

**** Academic Integrity and Regulations**

- All submitted work must be **original**.
- Plagiarism will result in disciplinary action.
- All external sources must be properly cited.
- Ethical handling of data is mandatory.
- Late submissions follow course policy.

**Final Remarks

- The project must result in a **working prototype solution**.
- A user interface and deployment are mandatory.
- Code-only or non-functional submissions will not be accepted.
- Clear problem understanding and practical implementation are key evaluation factors.